

PROGRAMME: BCA (Bachelor of Computer Applications)- MACT SEMESTER – V

Teaching-Learning & Evaluation Plan

Course Information: SKILL ENHANCEMENT COURSE -2

Course Code: 23BCA5CTE01 Course Title: MANAGEMENT OF DATA CENTERS

Credits Units: 03 Contact Hours + Experiential Hours: 45 + 45 L-T-P-E: 3-0-0-3

IA: ESE Weightage - 50: 50 Pass Marks (IA & ESE) – NA: 18 Aggregate Pass Marks: 40

ESE Question Paper Marks: 50 Special Examination Fees: NA

Pre-requisite (if any):

Course Facilitator (s): Prof. Diwakara Vasuman M S

Programme Outcomes (POs)	
At the end of the programme, students will be able to	
PO 1	Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.
PO 2	Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.
PO 3	Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand, and propose integrated solutions using emerging technologies.
PO 4	Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well informed conclusions.
PO 5	Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions.
PO 6	Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.
PO 7	Life-long Learning: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.

PO 8	Project Management: Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.
PO 9	Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.
PO 10	Societal & Environmental Concern: Ability to recognize economic, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.
PO 11	Individual & Team Work: Ability to work as a member or leader in diverse teams in multidisciplinary environment.



PO 12	Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.
Program Specific Outcomes (PSO's)	
PSO 1	Explore the impact of Data Centre in a global scenario and be a successful computer professional
PSO 2	Apply the knowledge of the data centers and design security mechanisms for the data center.
PSO 3	Recognise the need to use different packages and hardware resources to design a given application.
PSO 4	Enterprise and develop high-end mobile applications by considering security aspects.

Course Objectives (COB's)

COB1: To learn the data centers, how they have evolved over the years, different facilities and their requirements

COB2: To learn different aspects that have to be considered while designing data center and various server farms and etc.

COB3: To manage Server Systems and Data Centres Infrastructure Management.

COB4: To Utilize the Storage, Bandwidth, Efficiency of systems and other resources for Data centre.

COB5: To Monitoring the Networks and Resources.

Course Outcomes (CO's)

At the end of the course, students will be able to

Sl. No.	Course Outcome	Description	Bloom's Taxonomy Level
1.	CO 1	Explain basic concepts of datacentre and architecture.	L2
2.	CO 2	Manage Server Systems and Data Centres Infrastructure Management.	L4
3.	CO 3	Utilize the Storage, Bandwidth, Efficiency of systems and other resources for Data centre.	L3
4.	CO 4	Examine and Monitor the Networks and Resources.	L6
5.	CO 5	Planning and Creation for Flexible resource allocation.	L5

Sl. No.	Course Outcome	Skill's	Pos	Bloom's Taxonomy Level	Assessment Tool
1.	CO 1	Understanding	PO1, PO2, PO3, PO7, PO8	L2	Model Making, Presentation/Seminar
2.	CO 2	Apply	P01, P02, P03, P04, P05, P08, P09, P011	L3, L4, L5	Model Making, Mini Project
3.	CO 3	Analyze	P01, P02, P03, P04, P08, P09, P011	L3, L4, L5, L6	Model Making, Mini Project,
4.	CO 4	Evaluation	P01, P02, P03, P04, P05, P08, P09, P011	L3, L4	Assignment
5.	CO 5	Creation	P01, P04, P05, P06, P07, P09, P010, P011, P012	L2	Report Writing on Security Tools / Presentation

CO-PO/PSO Mapping:

Course Outcome	Blooms Taxonomy Levels	Program Outcomes (PO's) Mapped												Program Specific Outcomes (PSO's)			
		PO 1	PO 2	PO3	P O4	P O5	P O6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO 12	PSO 1 L2	PSO2 L3	PSO 3 L3	PSO 4 L6
CO1	L2	3	3	2	-	-	-	1	1	-	-	-	.	-	-	-	-

CO2	L4	3	3	2	3	3	-	-	-	1	-	-	2-	-	-	-	-
CO3	L3	2	3	3	2	3-	-	-	3	3	1	2	-	1	-	-	-
CO4	L6	2	3	3	3	3	3	-	3	2	-	2	-	-	-	-	1
CO5	L5	3	-	-	3	-	3	3	-	2	2	2	3	-	-	-	-
13	12	10	11	11	6	4	7	8	3	6	5	1	0	0	1	13	12



Course Contents:

Module	Details	Contact Hours
I	Overview of Data Centers Data Centres Defined, Data Centre Goals, Data Centre Facilities, Roles of Data Centres in the Enterprise, Roles of Data Centres in the Service Provider Environment, Application Architecture Models. The Client/Server Model and Its Evolution, The n Tier Model, Multitier Architecture Application Environment, Data Centre Architecture.	9
II	Data Centre Requirements Data Centre Prerequisites, Required Physical Area for Equipment and Unoccupied Space, Required Power to Run All the Devices, Required Cooling and HVAC, Required Weight, Required Network Bandwidth, Budget Constraints, selecting a Geographic Location, Safe from Natural Hazards, Safe from Man-Made Disasters, Availability of Local Technical Talent, Abundant and Inexpensive Utilities Such as Power and Water, Selecting an Existing Building (Retrofitting), tier standard.	9
II	Data Centre Design Characteristics of an Outstanding Design, Guidelines for Planning a Data Centre, Data Centre Structures, No-Raised or Raised Floor, Aisles, Ramp, Compulsory Local Building Codes, Raised Floor Design and Deployment, Plenum, Floor Tiles, Equipment Weight and Tile Strength, Electrical Wireways, Cable Trays, Design and Plan against Vandalism.	9
IV	Introduction to Server Farms Types of server farms and data centre, internet server farm, intranet server farm, extranet server farm, internet data centre, corporate data centre, software defined data centre, data centre topologies, Aggregation Layer, Access Layer, Front-End Segment, Application Segment, Back-End Segment, Storage Layer, Data Centre Transport Layer, Data Centre Services, IP Infrastructure Services, Application Services.	9
V	Business Continuity and Disaster Recovery fundamentals Business continuance infrastructure services, the need for redundancy, Information availability, BC terminology, BC planning life cycle, BC technology	9

	solutions, backup and recovery considerations, backup technologies, Uses of local replicas, Local replication technologies, Restore and restart considerations, Modes of remote replications, remote replication technologies.	
--	--	--



Text Books (TB):

TB-1	Data Centre Handbook 1st Edition by Hwaiyu Geng, 2020
TB-2	Requirements: Specifying, Designing, Building and Migrating to New Data Centers Kindle Edition by Art Carapol, 2018
TB-3	Data Center for Beginners: A beginner's guide towards understanding Data Center Design (Data Center Design Guide) Kindle Edition, 2017
TB-4	Guide To Planning, Designing and Building a Cloud Data Center 1st Edition by Caesar Wu (Author), Rajkumar Buyya , 2015

Reference Books (RB):

RB-1	Data Center Management: Your guide to efficient Data Center operation: 2 Paperback – Import, 31 July 2019
RB-2	Data Center Handbook Hardcover – 9 January 2015 by Hwaiyu Gen
RB-3	The Art of the Data Center 1st Edition by Douglas Alger , 2012

Other Reading (OR):

OR-1	Data Center - https://www.sunbirdcim.com/what-is-data-center-management
OR-2	The 3 Layers of Data Center Management - https://www.datacenterknowledge.com/archives/2011/07/05/the-3-layers-of-data-center-management
OR-3	Data Center Management

	https://www.nlyte.com/faqs/what-is-data-center-management/
OR-4	Business continuity and disaster recovery planning:- https://www.csoonline.com/article/2118605/business-continuity-and-disaster-



	recovery-planning-the-basics.html
--	--

Relevant Website (RW):

RW 1	https://www.chinatelecomglobal.com/
RW 2	https://www.ntt.com/en/index.html
RW 3	Certificate in Data center design - https://buildings.honeywell.com/us/en/lp/reliability-and-uptime?utm_campaign=datacenterSEM2021&utm_medium=paid-search&utm_source=google&utm_content=reliability_and_uptime&s_kwcid=AL!7892!3!502357098716!p!!g!!data%20centre%20management&gclid=CjwKCAjwzruGBhBAEiwAUqMR8K9LTG9WHxMEyKDlODfU7eBf7gTqStCYx7XdQ9zWiywxxi0h8N9TBoC8AcQAvD_BwE



Video Links (VL):

VL-1	Overview of Data Centers https://www.youtube.com/watch?v=FD5cf7uZDhg
VL-2	Data Center Infrastructure Design Webinar IEEE - https://www.youtube.com/watch?v=BzlvVBxSEOM
VL-3	Data Center HVAC - Cooling systems - https://www.youtube.com/watch?v=xBxyhxmhigc
VL-4	How to Perform Network Forensic https://www.youtube.com/watch?v=Ekp1jCUL4o0
VL-5	Server Farms https://www.youtube.com/watch?v=OpV6Yv0mzYk
VL-6	Business Continuity and Disaster Recovery fundamentals

Mapped Online Courses (OC):

Sr.No.	Platform	Topic	CO	Link	Duration
OC-1	Online Learning Udemey	Data Center Essentials: General Introduction	C01	https://www.udemy.com/course/data-center-essentials/	2 hours 30 min
OC-2	LinkedIn Learning	CompTIA Security+ (SY0- 501) Cert Prep: 3 Architecture and Design	C01 C03	https://www.linkedin.com/learning/comptia-security-plus-sy0-501-cert-prep-3-architecture-and-design/welcome?u=92695330	1 hour 52 min
OC-3	LinkedIn Learning	Cisco Network Security: Core Security Concepts	C03 C04	Defending the network (HIYA) - Cisco Video Tutorial LinkedIn Learning, formerly Lynda.com	1 hour 46 min
OC-4	LinkedIn Learning	AWS Administration: Security Fundamental	C03 C04	Fundamentals of AWS security - Amazon Web Services Video Tutorial LinkedIn Learning, formerly Lynda.com	2 hours 34 min
OC-5	LinkedIn Learning	Azure for Architects: Design a Business Continuity Strategy	C01 C03 C04 C05	https://www.linkedin.com/learning/azure-for-architects-design-a-business-continuity-strategy-2/design-your-business-continuity-and-disaster-recovery-strategy?u=92695330	1 hours 7 min
OC-6	LinkedIn Learning	Cloud data centers	C01 C03 C04 C05	https://www.linkedin.com/learning/search?keyword=data%20center&u=92695330	2 hours 42 min

Total MOOC Course integration with Certification
**12
hours**

Hardware/Software/Equipment (HSE):

HSE-1	Linux (Kali/Fedora)
HSE-2	Network Simulators
HSE-3	VMWare Workstation
HSE-4	ESXI Server
HSE-5	Router
HSE-6	Switch
HSE-7	Nagios, Open Source Software/learning website
HSE-8	Ganglia, Open Source Software/learning website
HSE-9	Untangle, Open Source Software/learning website
HSE-10	ClearOS Open Source Software/learning website

Abbreviations & Expansions

Abbreviations & Expansions			
Pedagogy/Activity planned		Mode of Delivery	
P1	PPT presentation & Classroom interaction	M1	Synchronous – PPT
P2	Blended Learning with Hands on	M2	Asynchronous / synchronous
P3	Flip Class & Module Quiz	M3	Synchronous - Hands on
A1	Activity-1	M4	Synchronous - Discussion
A2	Activity-2		

Session-wise Planning:

Module	Session	Topic	Links	TB/RB/OR /RW /VL	Pedagogy/ Activity planned	CO	Mode of delivery	Remarks
Bridge Course	Zero Lecture	Data Centres Defined	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	VL-1 to VL 5	P2	CO 1	Synchronous-PPT Asynchronous- Video	
	1	Data Centre Goals	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 OR1 VL1	P1		Synchronous-PPT Asynchronous- Video	
	2	Forms of Cyber Crime	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 OR1	P2		Interactive Lecture Synchronous-Video	
	3	Data Centre Facilities	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-4 OR2 VL1	PPT		Synchronous-PPT Asynchronous- Video	
	4	Roles of Data Centres in the Enterprise	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 OR2	P1		Synchronous-PPT Asynchronous- Video	
	5	Roles of Data Centres in the Service Provider Environment	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-2 OR1	P1		Synchronous-PPT Asynchronous- Video	
	6	Application Architecture Models	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 OR2	P3		Synchronous-PPT Asynchronous- Video	
	7	The Client/Server Model and Its Evolution	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	OR-2 RB-3	P1		Synchronous-PPT Asynchronous- Video	
	8	The n-Tier Model, Multi Tier Architecture Application Environment, ent	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-4	P1		Synchronous- Discussion, Suggestions	

	9	Activity-1- MOOC Course on Data Center Essentials: General Introduction	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	MC-1	P1		Assessment Certification on Completion	
	10	Data Centre Architecture: Internet edge connectivity	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 OR2	P1		Synchronous-PPT Asynchronous- Video	
	11	Server farms: front end segment, application layer, backend segm	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	VL-5	P1		Synchronous-PPT Asynchronous- Video	
	12	Case Study- Data Center Case Studies available at https://www.missioncriticalmagazine.com/top-ics/2907-data-center case-studies	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	OR-2 RB-3	P1		Assessment	
2	13	Need for data centre prerequisites, Data Centre Prerequisites	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-2 RB-1 OR-1 VL-2	P3	CO 2	Synchronous-PPT Asynchronous- Video	
	14	Required Physical Area for Equipment and Unoccupied Space	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-2 RB-1 OR-1 VL-2 VL3	P3		Synchronous-PPT Asynchronous- Video	
	15	Required Power to Run All the Devices	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 OR-1	P1		Interactive Lecture Synchronous-Video	
	16	Required Cooling and HVAC	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 OR-1	P1		Synchronous-PPT Asynchronous- Video	
	17	Required Weight, Required Network Bandwidth	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 OR-1	P1		Synchronous-PPT Asynchronous- Video	
	18	Budget Constraints, Types of budgets	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-2 RB-1 OR-1 VL3	P2		Synchronous-PPT Asynchronous- Video	
	19		https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	RW-4	P1		Synchron	

		Center hosting and Colocation Business [https://www.manageengine.com/network-monitoring/erpsuites/opmanager-cs.html]	om/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	RW-5 RW-6 MC-5			ous-PPT Asynchronous- Video	
	20	Selecting a Geographic Location	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 RB-3 OR-1 VL-2 MC5	P1		Synchronous-PPT Asynchronous- Video	
	21	Safe from Natural Hazards	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 RB-3 OR-1 VL-2 MC5	P1		Synchronous-Discussion, Suggestions	
	22	Safe from Man-Made Disasters	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 RB-3 OR-1 VL-2 MC5	P1		Assessment Certification on Completion	
	23	Availability of Local Technical Talent	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 RB-3 OR-1 VL-2 MC2	P1		Synchronous-PPT Asynchronous- Video	
	24	Mini Project with MOOC CompTIA Security+ (SY0-501) Cert Prep: 3 Architecture and Design MC2	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	RW-4 RW-5 RW-6 MC-5	P1		Synchronous-PPT Asynchronous- Video	
3	25	What is datacentre design? Characteristics of an Outstanding Design	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-4 RW-1 VL-4 MC-4	P1	C O 3	Synchronous-PPT Asynchronous- Video	
	26	Guidelines for Planning a Data Center	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-4 RW-1 VL-4 MC-4	P1		Synchronous-PPT Asynchronous- Video	
	27	Data Centre Structures, No-Raised or Raised Floor	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-4 RW-1 VL-4 MC-4	P1		Interactive Lecture Synchronous-Video	
	28	Difference between raised and no-raised floor	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-4 RW-1 VL-4 MC-4	P1		Synchronous-PPT Asynchronous- Video	
	29	Aisles, Ramp, Compulsory Local Building Codes	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-4 RW-1 VL-4 MC-4	P1		Synchronous-PPT Asynchronous- Video	

	30	Raised Floor Design and Deployment	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-4 RW-1 VL-4 MC-4	P1		Synchronous-PPT Asynchronous- Video	
	31	Plenum, Floor Tiles, Equipment Weight and Tile Strength	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-4 RW-1 VL-4 MC-4	P1		Synchronous-PPT Asynchronous- Video	
	32	Plenum, Floor Tiles, Equipment Weight and Tile Strength	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 VL-4	P2		Synchronous-PPT Asynchronous- Video	
	33	Activity-1- MOOC Course on Azure for Architects: Design a Business Continuity Strategy	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	MC-5	P1t		Synchronous-Discussion, Suggestions	
	34	Activity 2. Mooc Cloud data centers	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	MC-6	P1		Assessment Certification on Completion	
	35	Activity-2 Mini Project with Case Study Review & Data Center Design Case Studies https://www.ipspaceline.com/Data_Center_Design_Case_Studies	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 VL-6	P1n		Synchronous-PPT Asynchronous- Video	
4	36	Server Farms, Types of server farms and data centre	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 RB-2 RW-1 VL-4	P1	C O 4	Synchronous-PPT Asynchronous- Video	
	37	internet server farm, intranet server farm, extranet server farm	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 RB-3 MC-5	P1		Synchronous-PPT Asynchronous- Video	
	38	internet data center, corporate data center	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1 VL-3	P1		Synchronous-PPT Asynchronous- Video	
	39	software defined data center, Data center topologies	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-2 RB-3 VL-2	P1		Interactive Lecture Synchronous-Video	
	40		https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GiIyCimomWCRQB?usp=sharing	TB-1	P1		Synchronous	

		Access Layer, Front-End Segment	om/drive/folders/1qfEbZRkqHBeiyONE-5GilyCimomWCRQB?usp=sharing	RB-3 OR-3 RW-2 VL-1		ous-PPT Asynchronous- Video	
	41	Application Segment, Back-End Segment	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GilyCimomWCRQB?usp=sharing	TB-1 RW-1 VL-5	P1	Synchronous-PPT Asynchronous- Video	
	42	Storage Layer, Data Center Transport Layer, Data Center Services: IP Infrastructure Services,	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GilyCimomWCRQB?usp=sharing	TB-1 RB-3 RW-1 OR-3 RW-2	P1	Synchronous-PPT Asynchronous- Video	
	43	Activity-1- MOOC Cisco Network Security: Core Security Concepts	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GilyCimomWCRQB?usp=sharing	MC-4	P1	Assessment	
	44	Services, Storage Services	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GilyCimomWCRQB?usp=sharing	TB-4 VL-3	P1	Synchronous-PPT Asynchronous- Video	
5	45	Business continuance infrastructure services, the need for redundancy	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GilyCimomWCRQB?usp=sharing	TB-1 RW-8	P1	Synchronous- Discussion, Suggestions	
	46	Information availability , BC terminology ,	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GilyCimomWCRQB?usp=sharing	TB-2 OR2 RW-2 MC-2	P1	Assessment Certification on Completion	
	47	BC planning life cycle , BC technology solutions	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GilyCimomWCRQB?usp=sharing	TB-1 RB-3 RW-1 VL-6 MC-5	P2	Synchronous-PPT Asynchronous- Video	
	48	Backup considerations	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GilyCimomWCRQB?usp=sharing	TB-3 OR-4 RW-5 VL-4	P2	Synchronous-PPT Asynchronous- Video	
	49	recovery considerations	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GilyCimomWCRQB?usp=sharing	TB-1 RW-1 RW-2	P1	Synchronous-PPT Asynchronous- Video	
	50	backup technologies	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GilyCimomWCRQB?usp=sharing	TB-1 RB-1 OR-2 RW-3 VL-5	P1	Synchronous-PPT Asynchronous- Video	
	51	Local Replicas:	https://drive.google.com/drive/folders/1qfEbZRkqHBeiyONE-5GilyCimomWCRQB?usp=sharing	TB-4	P1	Synchronous	

		Replicas	om/drive/folders/ 1qfEbZRkqHBeiyO NE- 5GiIyCimomWCRQ B?usp=sharing	RB-3			ous-PPT Asynchro nous- Video	
	52	Uses of local replicas , Local replication technologies	https://drive.google.c om/drive/folders/ 1qfEbZRkqHBeiyO NE- 5GiIyCimomWCRQ B?usp=sharing	TB-4 RW-2 VL-6	P1n		Synchron ous-PPT Asynchro nous- Video	
	53	Restore and restart considerations	https://drive.google.c om/drive/folders/ 1qfEbZRkqHBeiyO NE- 5GiIyCimomWCRQ B?usp=sharing	TB-4 RW-2 VL-6	P1		Synchron ous-PPT Asynchro nous- Video	
	54	Modes of remote replications	https://drive.google.c om/drive/folders/ 1qfEbZRkqHBeiyO NE- 5GiIyCimomWCRQ B?usp=sharing	TB-2 RW-3 VL-6	P1		Synchron ous-PPT Asynchro nous- Video	
	55	MOOC Azure for Architects: Design a Business Continuity Strategy	https://drive.google.c om/drive/folders/ 1qfEbZRkqHBeiyO NE- 5GiIyCimomWCRQ B?usp=sharing	RW-4 RW-5 RW-6 MC-5	P2		Assessme nt	
	56	MOOC Azure for Architects: Design a Business Continuity Strategy	https://drive.google.c om/drive/folders/ 1qfEbZRkqHBeiyO NE- 5GiIyCimomWCRQ B?usp=sharing	RW-4 RW-5 RW-6 MC-5	P2		Assessme nt	
	57	MOOC Azure for Architects: Design a Business Continuity Strategy	https://drive.google.c om/drive/folders/ 1qfEbZRkqHBeiyO NE- 5GiIyCimomWCRQ B?usp=sharing	RW-4 RW-5 RW-6 MC-5	P1		Assessme nt	
	58	remote replication technologies	https://drive.google.c om/drive/folders/ 1qfEbZRkqHBeiyO NE- 5GiIyCimomWCRQ B?usp=sharing	RW-5	P2		Synchron ous-PPT Asynchro nous- Video	
	59	Recent Cyber Crime Case - Result Analysis	https://drive.google.c om/drive/folders/ 1qfEbZRkqHBeiyO NE- 5GiIyCimomWCRQ B?usp=sharing	RW-6	P1		Assessme nt	
	60	Certification with MOOC AWS Administration: Security Fundamentals	https://drive.google.c om/drive/folders/ 1qfEbZRkqHBeiyO NE- 5GiIyCimomWCRQ B?usp=sharing	MC-5	P2		Assessme nt	

Assessment Scheme: IA: UE - 50:50

Sl.	Assessment	Formative/	Frequency	Weightage	CO
-----	------------	------------	-----------	-----------	----

No.	Instrument	Summative		(%)	
1.	Activity 1-MOOC	Formative	1	15	CO1,CO2,CO3,CO4,CO5
2.	Activity 2-Mini Project		1	15	CO1,CO2,CO3,CO4,CO5
3.	Internal Test1		1	7.5	CO1, CO2,CO3
4.	Internal Test2	Formative	1	7.5	CO3,CO4,CO5
5.	Class participation		continuous	5	**5 MARKS based on class participation, activities submission on time and test attendance
6.	End Semester Exam	Summative	1	50	CO1, CO2, CO3, CO4,CO5
	Total			100	

Rubric for Activity-1 [MOOC Learning]

Criteria	Poor Below Expectation	Fair Needs Improvement	Good Meets Expectation	Excellent Exceeds Expectation	Weightage
Certificate Completion	(5-6 marks) Completed less than 15 hrs in 1-2 days	(7-8 marks) Completed 15 hrs or less in 2-3 days	(9 marks) Completed required 15 hrs in less 5-6 days	(10 marks) Completed required 15 hrs in 7-8 days	10
Report Submission	(2 marks) Poor Clarity in lessons	(3 marks) Moderate clarity in	(4 marks) Good Clarity in lessons	(5 marks) Very Good Clarity in lessons learned	5

	learned; poorly organised report	lessons learned & report	learned & organised report	& well organised report	
Oral Presentation	(2 marks) Poor language, voice, & Content	(3 marks) Moderate language, voice, & Content	(4 marks) Good language, voice, & Content	(5 marks) Very Good language, voice, & Content	5
Question Answer	(2 marks) Answers less than 3 questions correctly	(3 marks) Answers 3 questions correctly	(4 marks) Answers 4 questions correctly	(5 marks) Answers all 5 questions correctly	5

Rubric for Activity-2 Mini Project

Criteria	Poor Below Expectation	Fair Needs Improvement	Good Meets Expectation	Excellent Exceeds Expectation	Weightage
Report Submission	(8 marks) Poor: Trivial Idea, No Clarity in Items presented & plagiarism	(10-11 marks) Moderate: Trivial Idea, No much Clarity in Items presented & No plagiarism	(12-13 marks) Good Report: Trivial Idea, Clarity in Items presented & No plagiarism	(14-15 marks) Very Good Report: Innovative Clear Idea, Clarity in Items presented & No plagiarism	15
Oral Presentation	(2 marks) Poor language, voice, & Content	(3 marks) Moderate language, voice, & Content	(4 marks) Good language, voice, & Content	(5 marks) Very Good language, voice, & Content	5
Question Answer	(2 marks) Answers less than 3 questions correctly	(3 marks) Answers 3 questions correctly	(4 marks) Answers 4 questions correctly	(5 marks) Answers all 5 questions correctly	5

Assessment Sheet with Rubrics for Grading & Evaluation

